Week 04

UNIVERSITAS AMIKOM YOGYAKARTA

PROJECT SCOPE

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Learning objective

- To understand the concept of Project Scope Management and its importance in controlling project boundaries.
- To identify and define project requirements based on stakeholder needs, systematically develop scope statements and Work Breakdown Structures (WBS), and manage scope changes (scope control) to prevent scope creep.
- To ensure that all project work is in accordance with established objectives and that deliverables are achieved accurately.

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...Scope Management...

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What is Project Scope Management?

- **Scope** refers to *all* the work involved in creating the products of the project and the processes used to create them.
- A **deliverable** is a product produced as part of a project, such as hardware or software, planning documents, or meeting minutes.
- Project scope management includes the processes involved in defining and controlling what is or is not included in a project.

Project Scope Management Processes

- **Scope planning**: Deciding how the scope will be defined, verified, and controlled.
- **Scope definition**: Reviewing the project charter and preliminary scope statement and adding more information as requirements are developed and change requests are approved.
- **Creating the WBS**: Subdividing the major project deliverables into smaller, more manageable components.
- **Scope verification**: Formalizing acceptance of the project scope.
- **Scope control**: Controlling changes to project scope.

Sample Project Charter

Project Title: Information Technology (IT) Upgrade Project

Project Start Date: March 4, 2007 **Projected Finish Date:** December 4, 2007

Project Manager: Kim Nguyen, 691-2784, *knguyen@course.com*

Project Objectives: Upgrade hardware and software for all employees (approximately 2,000) within nine months based on new corporate standards. See attached sheet describing the new standards. Upgrades may affect servers, as well as associated network hardware and software. Budgeted $1,000,000 for hardware and software costs and $500,000 for labor costs.

Approach:

- Update the information technology inventory database to determine upgrade needs
- Develop detailed cost estimate for project and report to CIO
- Issue a request for quote to obtain hardware and software
- Use internal staff as much as possible for planning, analysis, and installation

ROLES AND RESPONSIBILITIES:

Name	*Role*	*Responsibility*
Walter Schmidt	CEO	Project sponsor, monitor project
Mike Zwack	CIO	Monitor project, provide staff
Kim Nguyen	Project Manager	Plan and execute project
Jeff Johnson	Director of Information, Technology Operations	Mentor Kim
Nancy Reynolds	VP, Human Resources	Provide staff, issue memo to all employees about project
Steve McCann	Director of Purchasing	Assist in purchasing hardware and software

Sample Project Charter (cont'd)

Sign-off: (Signatures of all the above stakeholders)

Walter Schmidt	*Steve McCann*
Mike Zwack	*Nancy Reynolds*
Kim Nguyen	*Jeff Johnson*

Comments: (Handwritten or typed comments from above stakeholders, if applicable)

"This project must be done within ten months at the absolute latest." Mike Zwack, CIO

"We are assuming that adequate staff will be available and committed to supporting this project. Some work must be done after hours to avoid work disruptions, and overtime will be provided." Jeff Johnson and Kim Nguyen, Information Technology department

Scope Definition and the Project Scope Statement

- The preliminary scope statement, project charter, organizational process assets, and approved change requests provide a basis for creating the project scope statement.

- As time progresses, the scope of a project should become clearer and more specific.

Further Defining Project Scope

Project Charter:

Upgrades may affect servers…

Preliminary Scope Statement:

Servers: If additional servers are required to support this project, they must be compatible with existing servers. If it is more economical to enhance existing servers, a detailed description of enhancements must be submitted to the CIO for approval. See current server specifications provided in Atch 6. The CEO must approve a detailed plan describing the servers and their location at least two weeks before installation.

Project Scope Statement, Version 1:

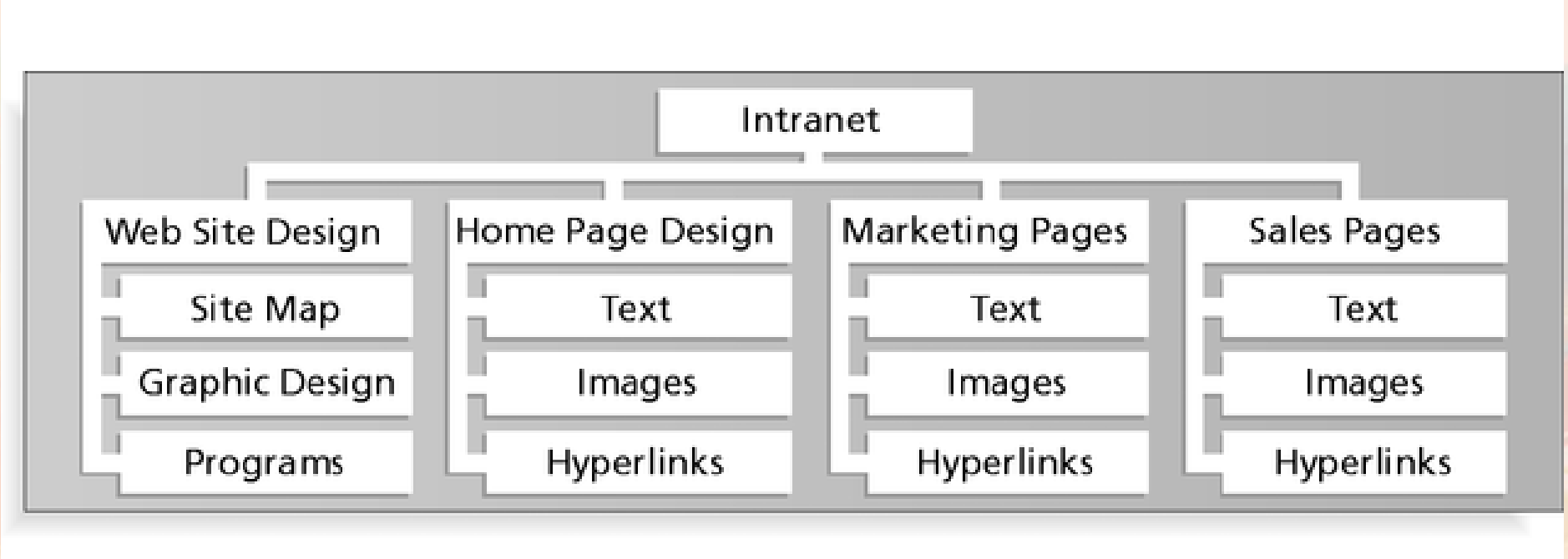
Servers: This project will require purchasing ten new servers to support Web, network, database, application, and printing functions. Two of each type of server will be purchased and dedicated to this project. Detailed descriptions of the servers are provided in a product brochure in Appendix 8 along with a plan describing where they will be located.

w	work
b	breakdown
s	structure

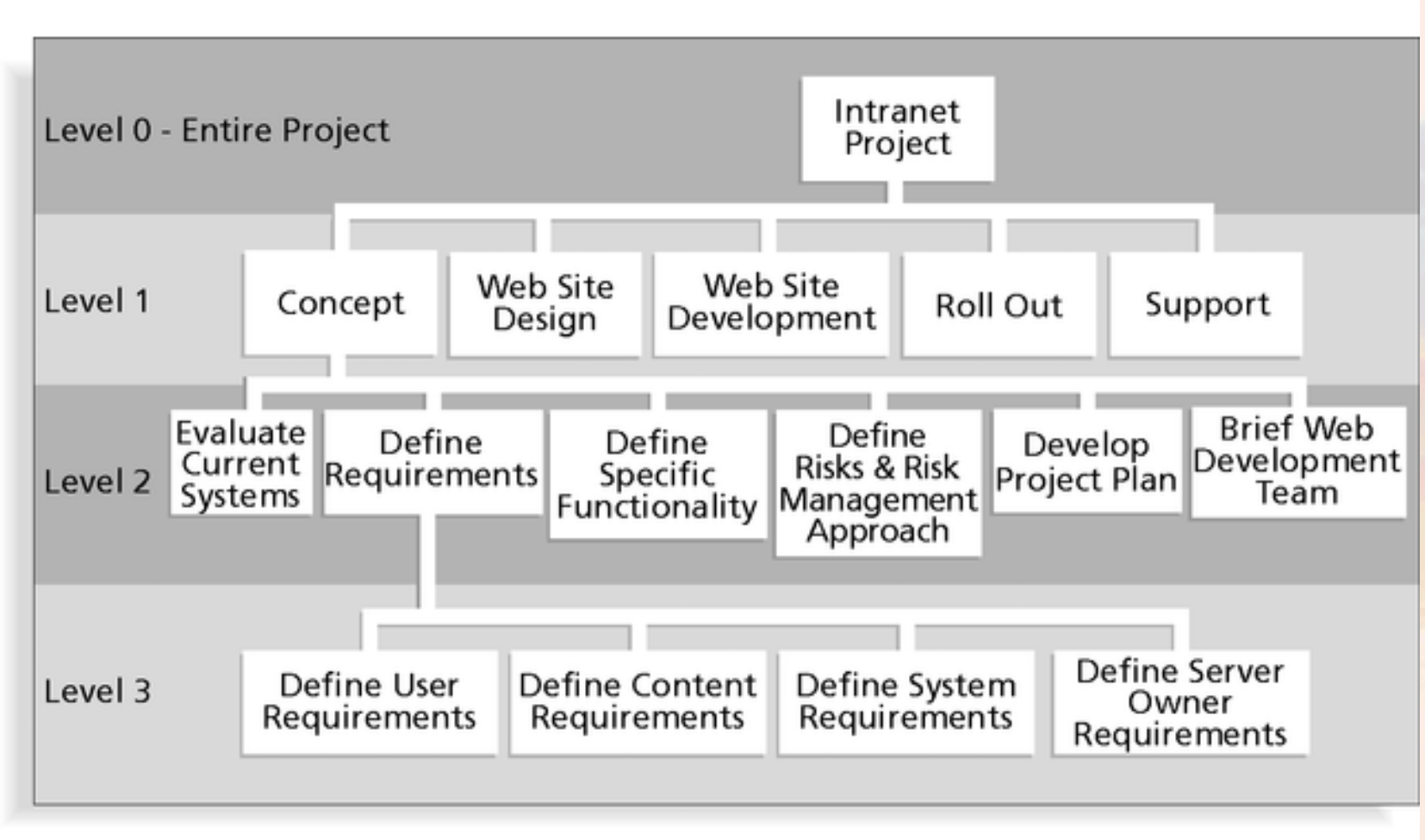
Creating the Work Breakdown Structure (WBS)

- A **WBS** is a deliverable-oriented grouping of the work involved in a project that defines the total scope of the project.
- A WBS is a foundation document that provides the basis for planning and managing project schedules, costs, resources, and changes.
- **Decomposition** is subdividing project deliverables into smaller pieces.

Sample Intranet WBS Organized by Product



Sample Intranet WBS Organized by Phase



Intranet WBS in Tabular Form

1.0 Concept
- 1.1 Evaluate current systems
- 1.2 Define requirements
  - 1.2.1 Define user requirements
  - 1.2.2 Define content requirements
  - 1.2.3 Define system requirements
  - 1.2.4 Define server owner requirements
- 1.3 Define specific functionality
- 1.4 Define risks and risk management approach
- 1.5 Develop project plan
- 1.6 Brief Web development team

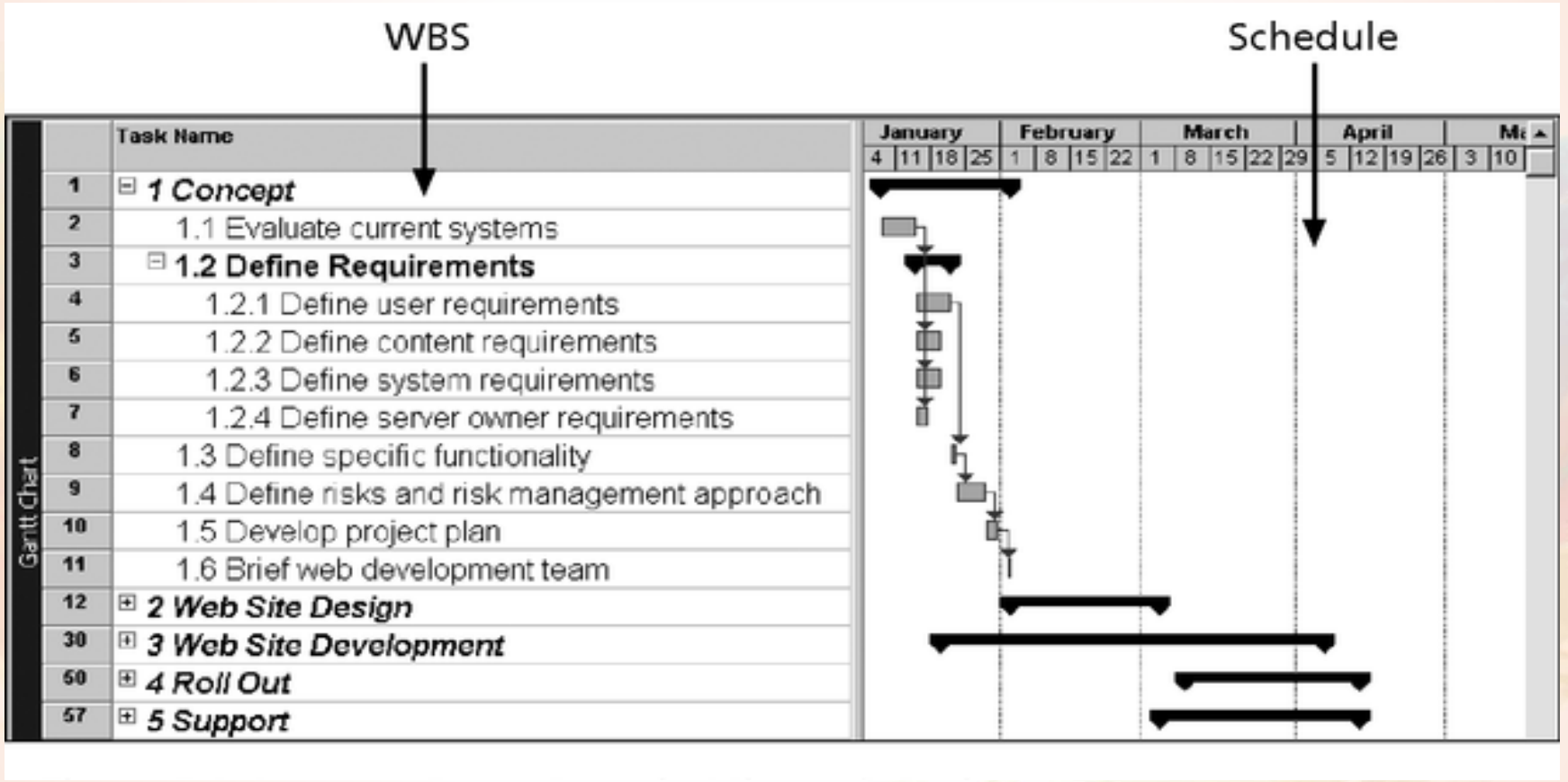
2.0 Web Site Design

3.0 Web Site Development

4.0 Roll Out

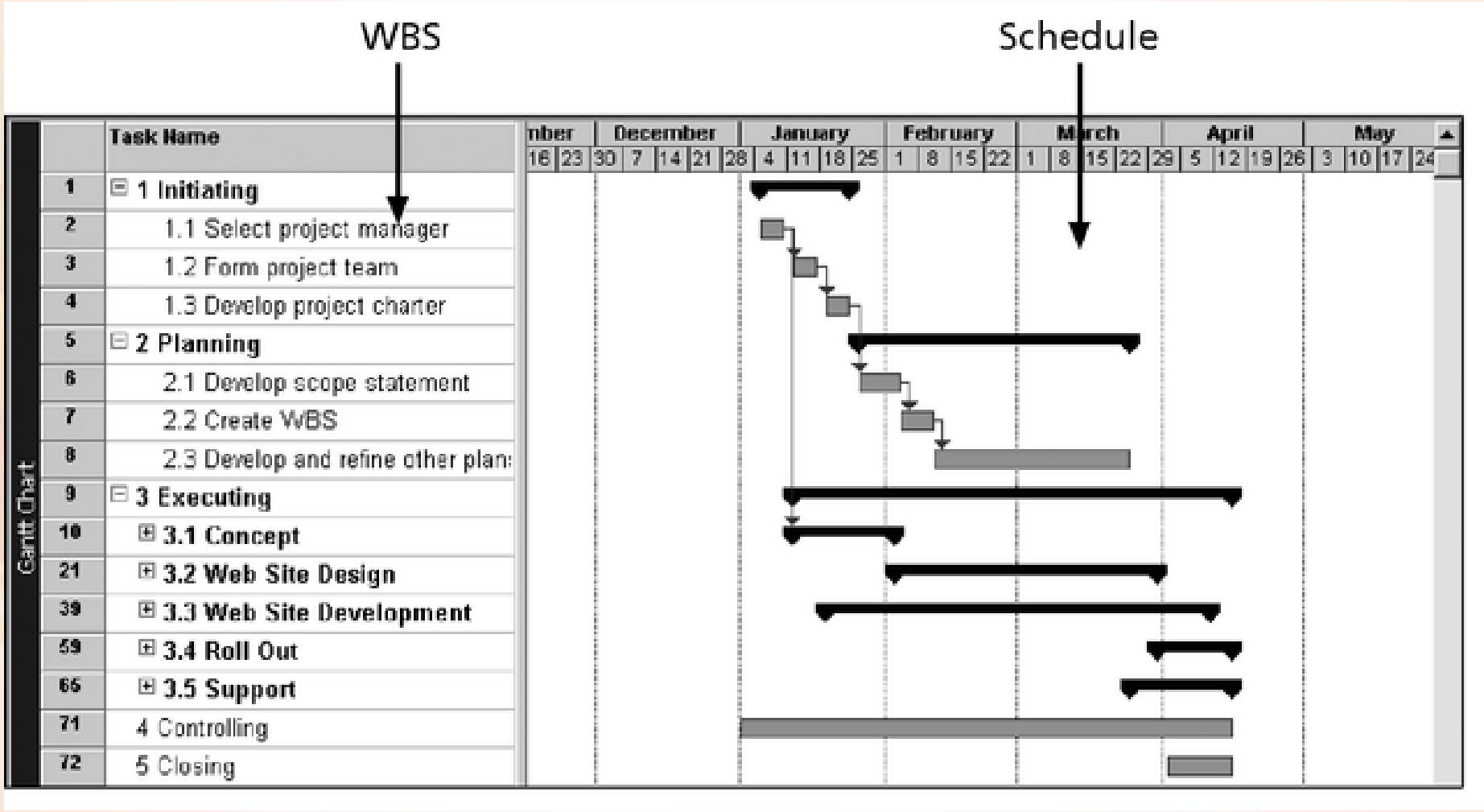
5.0 Support

Intranet WBS and Gantt Chart in Project 2000



Project 98 file

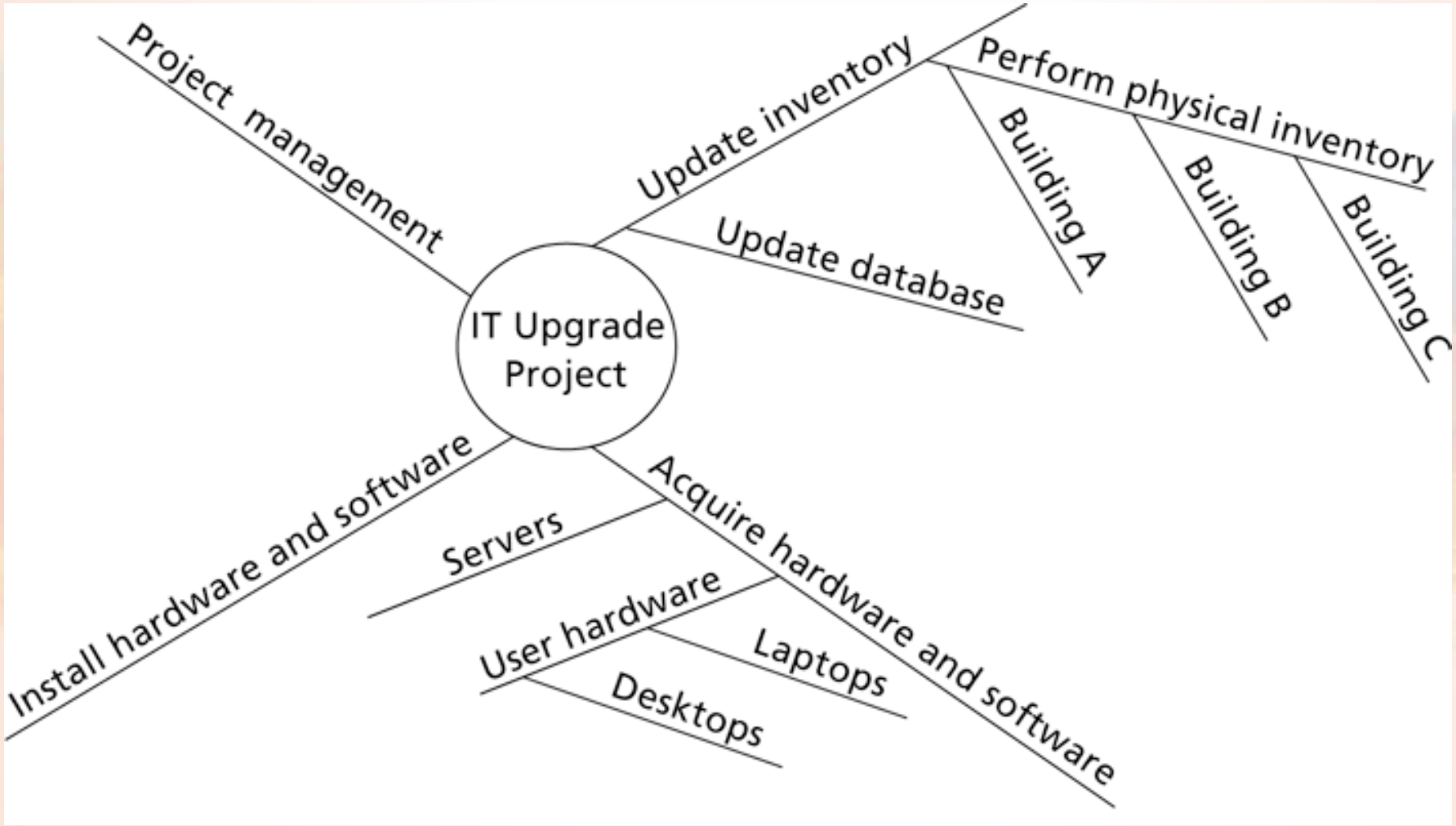
Intranet Gantt Chart Organized by Project Management Process Groups



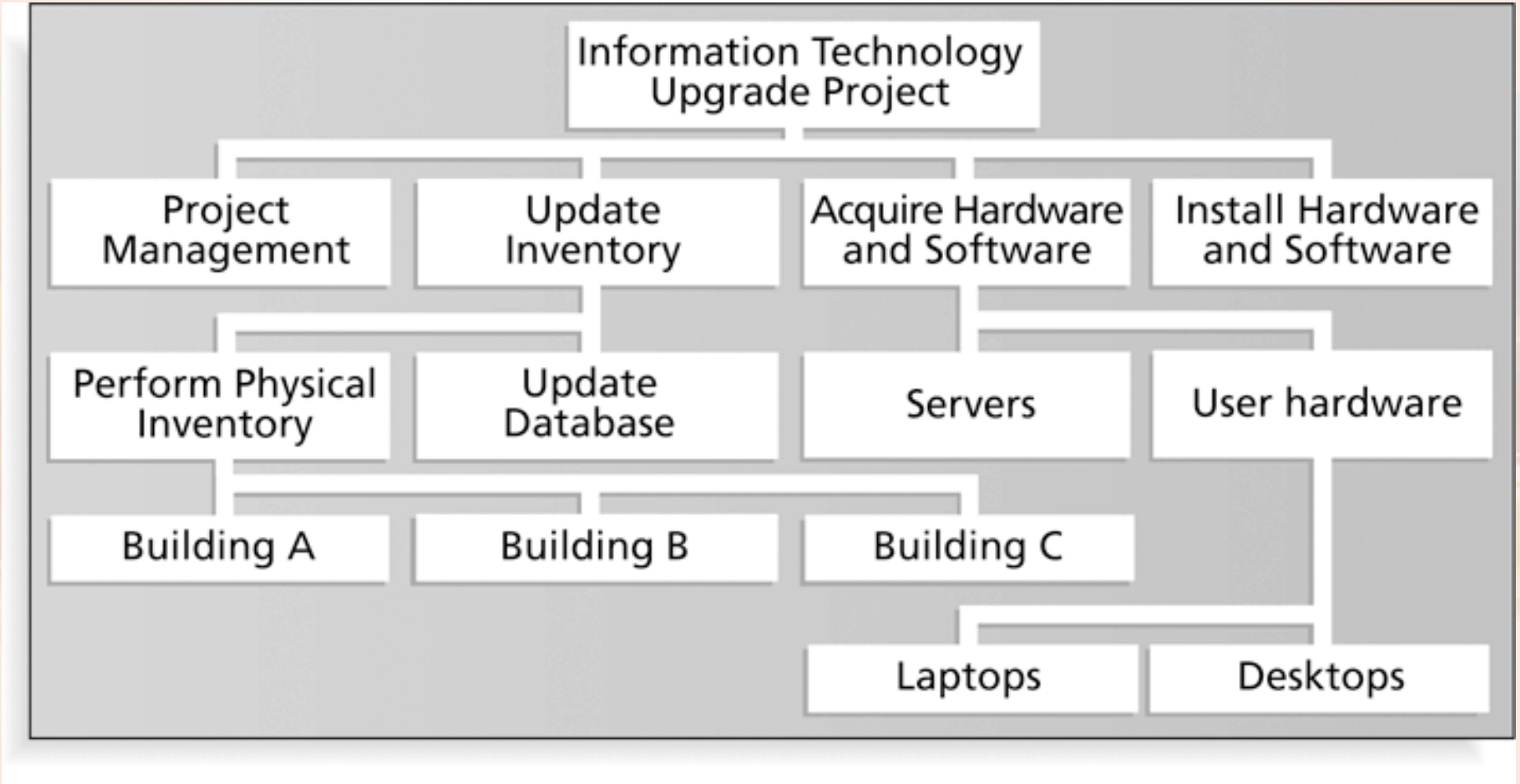
Approaches to Developing WBSs

- **Guidelines**: Some organizations, such as the DOD, provide guidelines for preparing WBSs.
- **Analogy approach**: Review WBSs of similar projects and tailor to your project.
- **Top-down approach**: Start with the largest items of the project and break them down.
- **Bottom-up approach**: Start with the specific tasks and roll them up.
- **Mind-mapping approach**: Write tasks in a non-linear, branching format and then create the WBS structure.

Sample Mind-Mapping Approach



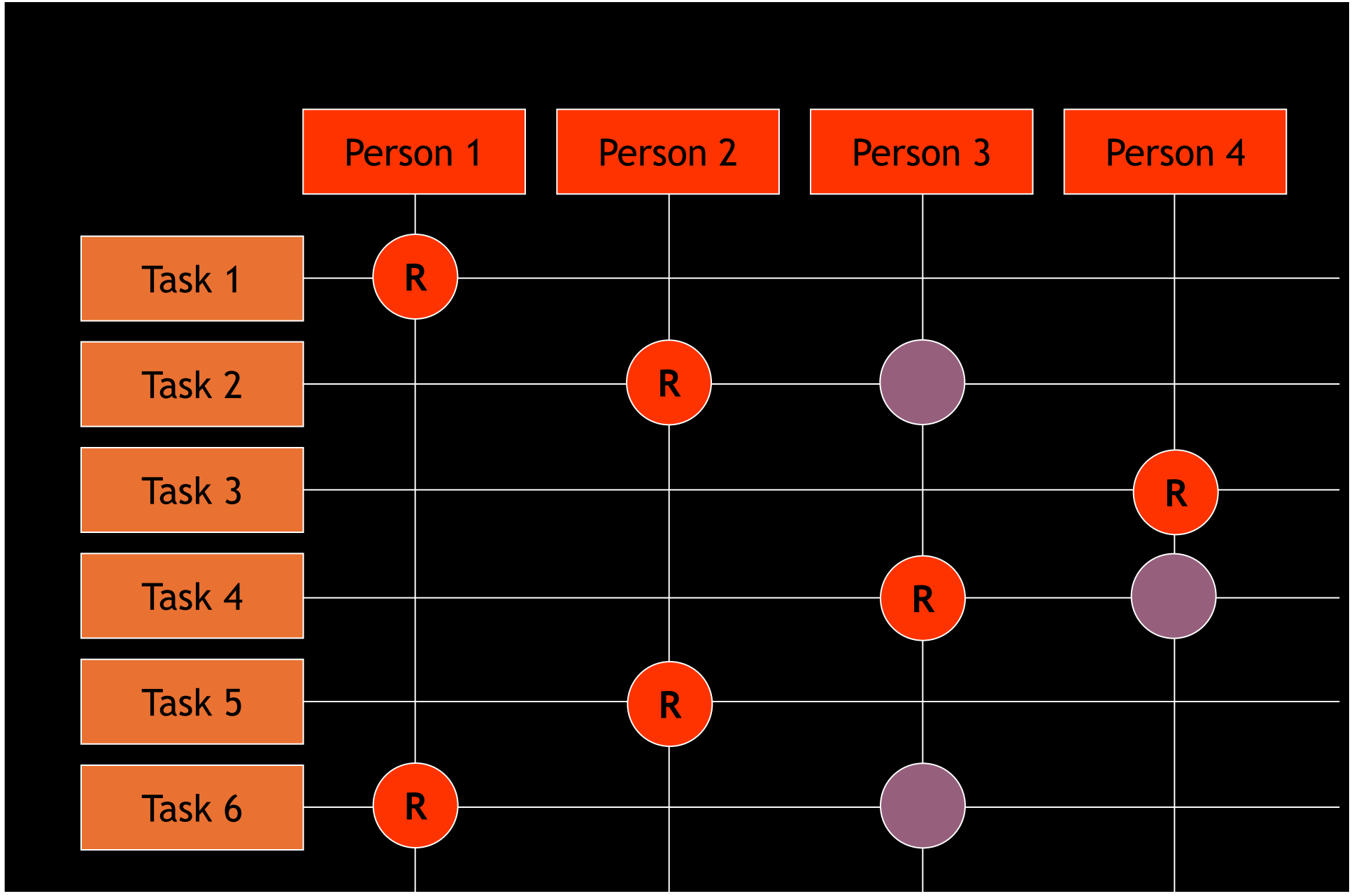
Resulting WBS in Chart Form



The WBS Dictionary and Scope Baseline

- Many WBS tasks are vague and must be explained in more detail so people know what to do and can estimate how long the work will take and what it will cost.
- A **WBS dictionary** is a document that describes detailed information about each WBS item.
- The approved project scope statement and its WBS and WBS dictionary form the **scope baseline**, which is used to measure performance in meeting project scope goals.

r	responsibility
a	assignment
m	matrix


Person 1
Person 2
Person 3
Person 4
Task 1
Task 2
Task 3
Task 4
Task 5
Task 6
R
R
R
R
R
R

Scope Verification

- It is very difficult to create a good scope statement and WBS for a project.
- It is even more difficult to verify project scope and minimize scope changes.
- Many IT projects suffer from scope creep and poor scope verification
  - FoxMeyer Drug filed for bankruptcy after scope creep on a robotic warehouse.
  - Engineers at Grumman called a system “Naziware” and refused to use it.
  - 21st Century Insurance Group wasted a lot of time and money on a project that could have used off-the-shelf components.

Scope Control

- **Scope control** involves controlling changes to the project scope.
- Goals of scope control are to:
  - Influence the factors that cause scope changes.
  - Ensure changes are processed according to procedures developed as part of integrated change control.
  - Manage changes when they occur.
- **Variance** is the difference between planned and actual performance.

Suggestions for Improving User Input

- Develop a good project selection process and insist that sponsors are from the user organization.
- Place users on the project team in important roles.
- Hold regular meetings with defined agendas, and have users sign off on key deliverables presented at meetings.
- Deliver something to users and sponsors on a regular basis.
- Don’t promise to deliver when you know you can’t.
- Co-locate users with developers.

Using Software to Assist in Project Scope Management

- Word-processing software helps create scope-related documents.
- Spreadsheets help perform financial calculations and weighed scoring models, and help develop charts and graphs.
- Communication software, such as e-mail and the Web, helps clarify and communicate scope information.
- Project management software helps create a WBS, the basis for tasks on a Gantt chart.
- Specialized software is available to assist in project scope management.